

Wireless and Mobile Networks - Theory Notes

1. Introduction to Wireless and Mobile Networks

Two Fundamental Challenges

The chapter addresses two distinct but interconnected challenges in wireless and mobile networking:

1. Wireless Communication Challenge: Making sure data can travel smoothly through the air (radio signals) instead of wires, which can be harder because of noise, interference, and weaker signals.

2. Mobility Challenge: Making sure a user can keep using the network even while moving without breaking the connection.

Important Distinction: Wireless does not always mean mobility. A device can be wireless (like a desktop with WiFi) but stationary, or mobile (like a laptop moving between wired connections) but not currently using wireless links.

2. Elements of Wireless Networks

Network Architecture Components

Understanding wireless network architecture requires examining four fundamental elements that work together to provide connectivity.

2.1 Wireless Hosts

Definition and Characteristics

A **wireless host** is simply a device that connects to the network without using wires.

- End devices include laptops, smartphones, tablets, and IoT devices.
- They run apps (like YouTube, WhatsApp) and follow network rules (protocols).

Types of Wireless Hosts

- **Stationary Wireless Hosts:** Devices that use wireless but stay in one place, like desktop PCs with WiFi or fixed IoT sensors.
- **Mobile Wireless Hosts:** Devices that use wireless and also move around, like smartphones, tablets, and laptops.
- **Application Diversity:** Wireless hosts support many kinds of applications, from simple sensor readings to heavy tasks like video streaming.

2.2 Base Stations

A base station is like a central hub that connects wireless devices to the larger network.

Function/Role:

A base station works as a **middle point** between wireless devices (like phones, laptops) and the wired Internet.

Main Jobs:

- Pass data back and forth between the Internet (wired network) and wireless devices.
- Manage the use of wireless signals in its area (so devices don't interfere with each other).
- Control which devices can connect (access and authentication).

Examples:

- **Cellular networks:** Cell towers, 4G (eNodeB), 5G (gNodeB).
- **Wi-Fi networks:** Wi-Fi access points (APs), wireless routers.
- **Coverage area:** Each base station serves a region called a **cell**.

2.3 Wireless Links

A wireless link is the connection between a wireless device (host) and a base station (or sometimes another device) that uses radio waves instead of cables.

Characteristics & Functions:

- Main way mobile devices (phones, laptops, etc.) connect to base stations.
- Can also act as **backbone links** (e.g., between towers or network elements).
- Needs **special sharing methods** (multiple access protocols) so many users can use the same link without interfering.

Key Features:

- **Transmission Rate:** Can be very slow (Kbps) or very fast (Gbps), depending on the technology (e.g., old 2G vs. 5G).
- **Transmission Distance:** Can cover just a few meters (Wi-Fi, Bluetooth) or several kilometers (cell towers).
- **Frequency Bands:** Different wireless systems use different parts of the radio spectrum.
- **Access Sharing:** Since the medium (air) is shared, coordination is important so devices don't "talk over" each other.

2.4 Network Infrastructure

Network infrastructure is the foundation of a computer network. It includes all the hardware, software, and systems that allow devices to connect, communicate, and share data.

Main Components:

- **Hardware:** Routers, switches, base stations, servers, cables, antennas, Wi-Fi access points.
- **Software/Protocols:** Operating systems, network management software, communication protocols (like TCP/IP).
- **Transmission Media:** Wired (copper, fiber optic) and wireless (radio waves, microwaves).

3. Wireless Network Operating Modes

3.1 Infrastructure Mode

Architecture and Operation: Infrastructure mode represents the most common deployment model for wireless networks, where base stations provide the bridge between wireless devices and wired networks.

Key Characteristics:

- Base stations connect mobile devices to the wired network infrastructure
- All communication flows through the base station
- Provides centralized control and management
- Enables seamless integration with existing Internet infrastructure

Handoff Process:

- Occurs when a mobile device changes from one base station to another
- Critical for maintaining ongoing connections during mobility
- Requires coordination between multiple base stations

Advantages:

- Reliable connectivity to broader Internet
- Centralized security and access control
- Better resource management and Quality of Service (QoS)

3.2 Ad Hoc Mode

Architecture and Operation: Ad hoc mode enables direct device-to-device communication without requiring infrastructure support.

Key Characteristics:

- No base stations required for operation
- Devices communicate directly with other devices within transmission range
- Self-organizing network topology
- Dynamic routing based on device positions and connectivity

Self-Organization Process:

- Nodes automatically discover neighboring devices
- Establish communication links based on proximity and signal strength
- Implement distributed routing protocols
- Adapt to topology changes as devices move

Limitations:

- Limited to devices within direct communication range
- No inherent connection to wider Internet infrastructure
- More complex routing and coordination protocols
- Potential security and reliability challenges

4. Wireless Network Taxonomy

Classification of Wireless Networks

Wireless networks can be grouped in two ways:

1. **Hop Count** → how many “hops” (steps) data takes from one device to another.
2. **Infrastructure** → whether the network needs fixed equipment (like base stations) or not.

1. Single-Hop Networks (data goes in one step)

- **With Infrastructure:**
 - **Examples:** Wi-Fi, Cellular (4G/5G).
 - **How it works:** Devices connect directly to a base station (router, cell tower).
 - **Internet access:** Yes, through the base station.
 - **Use case:** Most common way to access the Internet (home Wi-Fi, mobile data).
- **Without Infrastructure:**
 - **Examples:** Bluetooth, Ad hoc networks.
 - **How it works:** Devices talk directly to each other (no base station).
 - **Internet access:** No.
 - **Use case:** Small, personal or temporary connections (sharing files, headphones, etc.).

2. Multi-Hop Networks (data goes through several devices before reaching the target)

- **With Infrastructure:**
 - **Example:** Mesh networks.
 - **How it works:** Devices can forward data for others until it reaches a base station.
 - **Internet access:** Yes, eventually connects to the Internet through infrastructure.
 - **Use case:** Expanding coverage in places where one base station isn't enough.
- **Without Infrastructure:**
 - **Examples:** MANET (Mobile Ad Hoc Network), VANET (Vehicular Ad Hoc Network).

- **How it works:** Devices connect in a chain, passing data from one to another. No base station.
- **Internet access:** No.
- **Use case:** Military, disaster recovery, or vehicle-to-vehicle communication.

5. Wireless Link Characteristics

Fundamental Differences from Wired Links

Wireless communication faces unique challenges that make it significantly more complex than wired communication.

5.1 Signal Degradation Factors

1. Path Loss:

Radio signals naturally get weaker as they travel through space or objects, and environmental factors like buildings, terrain, or weather make the signal weaker, limiting communication range and reducing link quality.

2. Interference:

Wireless devices often share the same frequency bands, and signals from other Wi-Fi networks, cellular networks, microwave ovens, or motors can interfere, causing packet loss, slower data speeds, and unstable connections.

3. Multipath Propagation:

Signals can bounce off buildings, vehicles, and other objects, so multiple copies of the same signal arrive at different times, causing distortion, fading, and errors, which need advanced signal processing to fix.

5.2 Signal-to-Noise Ratio (SNR) Considerations

Definition & Importance:

- SNR is the ratio of the signal's power to the background noise power.
- Higher SNR means a **cleaner, clearer signal**.
- It directly affects how fast and reliably data can be transmitted.

SNR vs. Bit Error Rate (BER):

- Higher SNR → fewer errors (lower BER).
- Different modulation methods need different SNR levels.
- There's a trade-off: faster data rates need higher SNR, while lower SNR may require slower but more reliable transmission.

Dynamic Adaptation Strategies:

- **Power Control:** Increase transmission power to improve SNR (limited by battery and regulations).
- **Modulation Adaptation:** Use more robust (simpler) modulation when SNR is low.

- **Rate Adaptation:** Reduce data rate to keep errors acceptable when signal is weak.
- **Mobility:** As devices move, SNR changes, so the system must continuously adjust.

5.3 Hidden Terminal Problem

Problem Description: The **Hidden Terminal Problem** occurs in **wireless networks** (like Wi-Fi) when **two devices cannot hear each other** but their transmissions can collide at a common receiver.

Scenario:

- There are three nodes: A, B, and C.
- Node B can hear both A and C, but A and C **cannot hear each other** because of distance or obstacles.
- If A and C try to send data to B at the same time, their signals **collide at B**, even though both checked the channel before transmitting.

Consequences:

- Collisions reduce **network throughput**.
- Some nodes may get **less chance to transmit** (unfair access).
- Collisions are **harder to detect** in wireless networks.
- Requires **special collision avoidance methods**.

Solutions / Mitigation:

- **RTS/CTS Protocol:** Nodes request permission before sending to avoid collisions.
- **CSMA/CA:**

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

A **medium access control (MAC) protocol** used in wireless networks (like Wi-Fi).

How it works:

1. A station first **listens to the channel** (carrier sense).
2. If the channel is **idle**, it waits a short random time (backoff) before sending data.
3. If the channel is **busy**, it waits until it's free.
4. Uses **ACK packets** to confirm successful reception (to avoid retransmission due to hidden terminal problem).

Purpose:

- Avoid collisions in **wireless networks** (where collision detection is hard).

Example:

- **IEEE 802.11 (Wi-Fi)** uses CSMA/CA.

6. Code Division Multiple Access (CDMA)

CDMA (Code Division Multiple Access)

Fundamental Idea:

CDMA lets **many users share the same frequency band at the same time** by giving each user a **unique code**. This allows multiple conversations to happen simultaneously without interfering much.

1. Basic CDMA Concepts

How it works:

- Multiple users transmit simultaneously over the same frequency band.
- Each user is assigned a unique code (spreading code).
- The receiver uses the same code to extract the intended signal while ignoring others.
- Signals overlap in time and frequency but remain distinguishable by their codes.

Purpose:

- Allow many users to share the same frequency without interfering.

Example:

- 3G mobile networks used CDMA.

7. IEEE 802.11 Wireless LAN (WiFi)

7.1 WiFi Standards Evolution

Historical Development and Technical Specifications:

The IEEE 802.11 family has evolved significantly since its inception, with each generation offering improved performance and capabilities.

IEEE 802.11 Wireless LAN

IEEE 802.11 standard	Year	Max data rate	Range	Frequency
802.11b	1999	11 Mbps	30 m	2.4 GHz
802.11g	2003	54 Mbps	30m	2.4 GHz
802.11n (WiFi 4)	2009	600	70m	2.4, 5 GHz
802.11ac (WiFi 5)	2013	3.47Gpbs	70m	5 GHz
802.11ax (WiFi 6)	2020 (exp.)	14 Gbps	70m	2.4, 5 GHz
802.11af	2014	35 – 560 Mbps	1 Km	unused TV bands (54-790 MHz)
802.11ah	2017	347Mbps	1 Km	900 Mhz

7.3 Wi-Fi Association Process

When a device wants to connect to a Wi-Fi network, it goes through the **association process**, which includes **finding an access point (AP)** and **establishing a connection**.

1. Channel Selection and Scanning

- Wi-Fi networks work on specific channels in a frequency band.
- The device scans to **find the best AP** to connect to.

1. Active Scanning

- **Definition:** The device (like a Wi-Fi client) **actively sends probe requests** to discover available networks.
- **How it works:**
 1. Client sends **probe requests** to all channels.
 2. Access Points (APs) that receive the request respond with **probe responses** containing their SSID and capabilities.
- **Characteristics:**
 - Faster discovery of networks.
 - Can reveal hidden networks (APs with hidden SSIDs).
 - Generates extra traffic in the network.
- **Example:** A smartphone searching for Wi-Fi networks will actively send requests and wait for responses from nearby APs.

2. Passive Scanning

- **Definition:** The device **listens for beacon frames** sent periodically by APs, without sending any requests.
- **How it works:**
 1. APs broadcast **beacon frames** regularly (every 100 ms typically) containing SSID, supported rates, and capabilities.
 2. Client simply **monitors these beacons** to find available networks.
- **Characteristics:**
 - Slower than active scanning.
 - Stealthier (does not generate network traffic).
 - Cannot detect APs that do not broadcast their SSID (hidden networks).
- **Example:** A Wi-Fi adapter in “monitor mode” listening to beacons to map networks without actively probing.

4. Post-Association Steps

- **Authentication:** Device proves it has the correct security credentials.
- **DHCP Configuration:** Device gets an IP address from the network.
- **Network Access:** Device now has full access to the local network and Internet.

7.4 IEEE 802.11 MAC Protocol: CSMA/CA

Purpose:

- Avoid collisions in wireless networks where devices share the same channel.
- Wireless devices **cannot detect collisions easily**, so they **avoid collisions before transmission**.

1. 802.11 Sender Operation

1. **Sense the channel:**
 - If the channel is **idle for DIFS** (Distributed Inter-Frame Space), transmit the entire frame.
2. **If the channel is busy:**
 - Start a **random backoff timer**.
 - Timer **counts down only when the channel is idle**.
 - When the timer expires, transmit the frame.
3. **If no ACK is received:**
 - Increase the random backoff interval (exponential backoff).
 - Repeat the process.

2. 802.11 Receiver Operation

1. **If frame is received correctly:**
 - Send **ACK** after **SIFS** (Short Inter-Frame Space).

3. Inter-Frame Spaces

Abbreviation	Meaning	Purpose
DIFS	Distributed Inter-Frame Space	Sender waits before transmitting data if channel is idle
SIFS	Short Inter-Frame Space	Receiver waits before sending ACK (priority over other traffic)

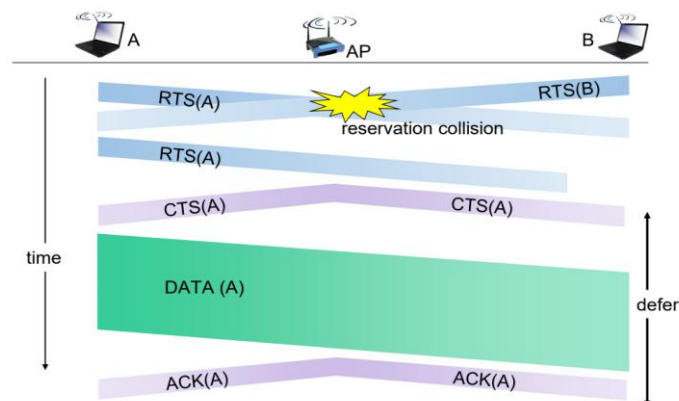
4. Collision Avoidance: RTS/CTS Exchange

- **Problem:** Hidden terminals can cause collisions.

Steps:

1. **RTS (Request-to-Send):**
 - Sender transmits a small RTS packet to the AP (or BS) using CSMA.
 - RTS packets may still collide, but they are **short**, so minimal waste.
2. **CTS (Clear-to-Send):**
 - AP responds with CTS to the sender.
 - **CTS is heard by all nearby nodes**, signaling them to **defer transmissions**.
3. **Data Transmission:**
 - Sender transmits the actual data frame.
 - Other stations **wait** until channel is free.

Collision Avoidance: RTS-CTS exchange

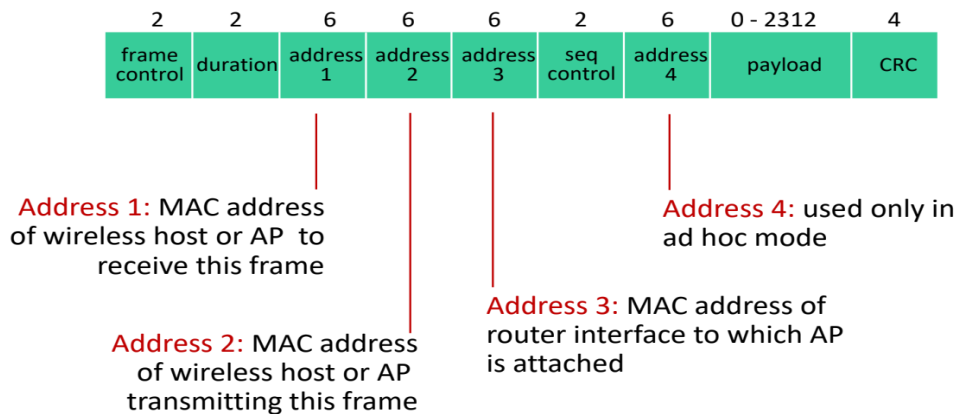


7.5 802.11 Frame Structure and Addressing

Frame Format Complexity: 802.11 frames are more complex than Ethernet frames due to the wireless environment and mobility requirements.

Key Frame Fields:

- **Frame Control (2 bytes):** Frame type, protocol version, control flags
- **Duration (2 bytes):** Channel reservation time for RTS/CTS
- **Address Fields (4 × 6 bytes):** Multiple addresses for complex routing scenarios
- **Sequence Control (2 bytes):** Frame sequence number for reliability
- **Payload (0-2312 bytes):** Actual data being transmitted
- **CRC (4 bytes):** Error detection code



Address Field Usage:

- **Address 1:** Destination MAC address (immediate recipient)
- **Address 2:** Source MAC address (immediate sender)
- **Address 3:** Router interface MAC address for infrastructure mode
- **Address 4:** Used only in ad hoc mode for complex routing

8. Personal Area Networks: Bluetooth

8.1 Bluetooth Architecture and Characteristics

Design Philosophy: Bluetooth was specifically designed as a short-range, low-power wireless technology to replace cables in personal area networks.

Piconet Architecture: Bluetooth networks form small clusters called piconets with a master-slave relationship structure.

Roles:

- **Master (M):** Controls the network, polls clients, grants transmission permissions.
- **Client (C):** Active device communicating with master.
- **Parked device (P):** Inactive, "sleep mode," but can wake up later to save battery.

Master Device Responsibilities:

- **Polling:** Master polls clients to grant transmission opportunities
- **Frequency Hopping Control:** Master determines frequency hopping sequence
- **Access Control:** Master grants transmission requests from clients
- **Timing Synchronization:** Master provides timing reference for the piconet

8.2 Bluetooth Medium Access and Frequency Management

9. Cellular Networks: 4G and 5G

9.1 Cellular Network Overview

Market Significance and Deployment: Cellular networks represent the dominant technology for wide-area mobile Internet connectivity, with massive global deployment and usage.

Technical Standards Evolution:

- **4G Standard:** Long-Term Evolution (LTE) specification
- **5G Standard:** New Radio (NR) specification with enhanced capabilities

9.2 Similarities & Differences with Wired Internet

Similarities

- Edge/core distinction (all under same carrier).
- Uses same Internet protocols: HTTP, DNS, TCP, UDP, IP, NAT.
- Separation of data/control planes (SDN style).
- Interconnected to wired Internet.

Differences

- Wireless link layer is different.
- **Mobility as core service** (handover between cells).
- User **identity via SIM card**.
- Subscription model: home vs roaming network.
- Global access with inter-carrier authentication & settlements.

9.3 4G LTE Architecture Components

User Equipment (UE)

- Smartphone, tablet, laptop, IoT device with LTE radio.
- Identity: IMSI (International Mobile Subscriber Identity) stored in SIM.

Base Station (eNode-B)

- At the network edge, like a Wi-Fi AP but with more responsibilities:
 - Manages radio resources, devices in coverage.
 - Handles user mobility (handover).

- Coordinates with neighbor base stations.

Core Components

- **MME (Mobility Management Entity)**

Main Functions

1. Authentication & Security

- When your phone connects, the MME checks your **SIM card (IMSI)** with the **HSS**.
- Ensures only valid subscribers can use the network.
- Helps set up **encryption** for secure communication.

2. Mobility Management

- **Handover:** If you move from one tower to another, MME makes sure your call/data continues without dropping.
- **Paging:** If someone calls you, MME finds your device by sending paging messages across possible towers.
- **Tracking:** Keeps track of which **cell area** your device is in (not exact location, but enough to deliver service).

3. Session Management

- Sets up the **tunneling path** for user data:
 - Phone → eNode-B → S-GW → P-GW → Internet.
- Maintains and updates this tunnel when you move.

- **HSS (Home Subscriber Server):**

- Stores subscriber info (identity, services).
- Works with MME for authentication.

- **S-GW (Serving Gateway):**

- Routes user traffic between eNode-B and P-GW.
- Supports mobility via tunneling.

- **P-GW (Packet Data Network Gateway):**

- Gateway to Internet.
- Provides NAT and routing.

10. Mobility Management Principles

10.1 Understanding Mobility from Network Perspective

Mobility means a device keeps moving across networks but still wants to stay connected (calls, internet, video streaming should not break).

10.2 Fundamental Mobility Approaches

1. Network-Based (Router-Centric)

- **Idea:** Let the **routers in the network** keep track of where a mobile is.
- **How it works:** Routers update their routing tables with the current location of mobiles and advertise it.
- **Problem:**
 - Too many mobiles (billions) → routing tables would become huge.
 - Updates would be very frequent → too much overhead.
 - Slow to update → not practical for fast-moving devices.
Not scalable for the real Internet.

2. End-System Based (Host-Centric)

Mobility is handled by **the mobile device itself** and the systems it talks to.

End-System Based (Host-Centric) Mobility Management is a mobility approach where the **mobile device (end host)** itself is responsible for managing its mobility, including updating its location, signaling, and handling handovers, instead of relying on the network.

How it Works

1. When the mobile host changes its point of attachment (e.g., moves to a new network), it **detects the movement**.
2. The mobile host obtains a **new IP address** (or care-of address).
3. It directly **updates its Home Agent (HA)** or the **Correspondent Node (CN)** with the new address.
4. Communication continues through the updated address without the network handling mobility on its behalf.

10.4 Home Network and Visited Network Concepts

1. Home Network

- Your home network is the main service provider you pay for, like Verizon, Orange, or T-Mobile.
- It has a Home Subscriber Server (HSS) that stores all your information: identity, subscriptions, and service profiles.
- You have a billing relationship with this provider, so they handle your payments and service plans.
- Your SIM card contains global identity information (IMSI) that links your device to your home network, no matter where you go.

2. Visited Network

- A visited network is any network you connect to that is not your home network.
- Your phone can connect through roaming agreements between your home network and the visited network.
- The visited network works with your home network to authenticate you and make sure you are allowed access.
- The visited network may offer limited services compared to what you normally get on your home network.

3. ISP/WiFi Case

- Unlike 4G/5G, **WiFi/ISP doesn't have a global "home" network.**
- Your device uses **credentials** (username/password) to connect.
- ISPs may exist nationally or internationally, but **each network may require separate credentials.**
- Some exceptions exist (like **eduroam**) where roaming is supported across networks.
- Mobile IP architecture exists to allow 4G-like mobility on WiFi, but it's **rarely used.**

4. Mobility with Indirect Routing

- Data from correspondent is sent to **home network first**, then forwarded (tunneled) to mobile.
- **Pros:** Correspondent doesn't need to know where mobile is → transparent.
- **Cons:** Inefficient → triangle routing (takes longer path).
- **Mobile moving across networks:** still works → ongoing connections (like TCP) continue without breaking.

5. Mobility with Direct Routing

- Correspondent **asks home network** for mobile's **current visited network address**.
- Sends data **directly** to mobile.
- **Pros:** Avoids triangle routing → more efficient.
- **Cons:** Correspondent must track mobile's current location → not transparent.
- If mobile moves again → needs update → extra complexity.